

Supply Chain Logistics Manager - Integration Architecture & A

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Version 1.0

1. System Context

The Logistics Manager platform integrates with carrier networks, ERP systems, and IoT devices to orchestrate global supply chain operations.

External Services: FedEx Ship API (shipping/tracking), UPS Developer Kit (shipping/tracking), DHL Express API (shipping/tracking), USPS Web Tools (shipping/tracking), Google Maps Platform (geocoding/routing), OpenWeatherMap (weather alerts), Avalara (duty/tax), C-TPAT/AEO (trade compliance).

Internal Services: Shipment Service (lifecycle management), Tracking Service (GPS/status aggregation), Warehouse Service (inventory/WMS), Routing Service (optimization engine), Customs Service (compliance/documents), Carrier Service (rate shopping/booking), Notification Service (email/SMS/webhook), Analytics Service (reporting/Bi).

2. Carrier API Integration

REQ-LOG-061: FedEx Ship API v27 integration: CreateShipment, RateRequest, TrackRequest, CancelShipment. OAuth 2.0 client credentials authentication.

REQ-LOG-062: UPS Developer Kit: Rating, Shipping, Tracking, Void APIs. OAuth 2.0 with refresh token rotation.

REQ-LOG-063: DHL Express API: Rate Request, Shipment Request, Tracking, Pickup. API key authentication with IP whitelisting.

REQ-LOG-064: Carrier webhook receivers for real-time tracking updates. Webhook signature verification using HMAC-SHA256.

REQ-LOG-065: Carrier abstraction layer: Normalized data model for shipment, rate, and tracking data regardless of carrier-specific formats.

3. ERP/OMS Integration

REQ-LOG-066: SAP S/4HANA integration via OData REST API for order and inventory synchronization.

REQ-LOG-067: Oracle NetSuite SuiteTalk API for mid-market ERP integration. Order import and fulfillment status export.

REQ-LOG-068: Shopify/BigCommerce webhook integration for e-commerce order fulfillment. Auto-creates shipments from new orders.

REQ-LOG-069: Generic EDI 856 (ASN) and EDI 214 (Transportation Status) for legacy ERP systems.

4. IoT & Sensor Integration

REQ-LOG-070: GPS tracker devices (CalAmp/Samsara) streaming location data via MQTT to AWS IoT Core.

REQ-LOG-071: Temperature/humidity sensors for cold chain shipments. Alert when temperature exceeds -18°C to -22°C range for frozen goods.

REQ-LOG-072: RFID readers at warehouse dock doors auto-detecting pallet-level inventory movements.

5. Sequence Flow: Rate Shopping & Booking

1. Web App -> Shipment Service: POST /api/shipments/rate-shop {origin, destination, packages, service_level}

2. Shipment Service -> Carrier Service: Fan-out rate requests to all integrated carriers in

parallel

3. Carrier Service -> FedEx API: POST /rate/v1/rates/quotes {shipment_details}
4. Carrier Service -> UPS API: POST /api/rating/v1/Shop {shipment_details}
5. Carrier Service -> DHL API: POST /rates {shipment_details}
6. All carriers respond within 3-second timeout. Non-responsive carriers excluded with note.
7. Carrier Service -> Shipment Service: Normalized rate responses [{carrier, service, transit_days, cost}]
8. Shipment Service -> Web App: Rate comparison table displayed to dispatcher
9. Dispatcher selects carrier -> Web App -> Shipment Service: POST /api/shipments/book {shipment_id, carrier, service}
10. Shipment Service -> Carrier Service -> Selected Carrier API: Create shipment, generate label
11. Carrier API responds: {tracking_number, label_url, pickup_confirmation}
12. Shipment Service -> Notification Service: Send tracking number to shipper and consignee